

Junkun Jiang

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EDUCATION

Southwest Jiaotong University - Traffic Engineering - Undergraduate 2023-09 ~ 2027-06

Campus Experience

- Minister, University Robotics Association — led R&D initiatives; a core team member representing the university in the National College Robotics Competition, achieving outstanding results
- Member, School of Traffic and Transportation Debate Team and College Basketball Team

RESEARCH EXPERIENCE

Tsinghua University, Institute for AI Industry Research (AIR) - SLAM Algorithm Research Intern 2026-07 ~ Present

- Started with pure inertial odometry (IO) — developing a learning-based phase-informed dynamic inertial odometry framework tailored to the unique oscillatory motion patterns of butterfly-inspired flapping-wing robots, aiming to address severe vibration-induced IMU degradation
- Progressed to visual-inertial odometry (VIO) by integrating monocular camera and event camera data, building a lightweight sensor fusion framework to explore robust state estimation under aggressive flapping motion, frequent lighting changes, and limited onboard compute
- The phase-informed approach leverages the periodic nature of wing flapping to disambiguate motion dynamics, with the goal of achieving more accurate pose estimation than conventional IMU-driven methods in high-vibration regimes

National College Students Robotics Competition : RoboMaster - Core R&D of Mapping and Positioning

2024-07 ~ 2025-07

- Developed a LiDAR-inertial SLAM system based on Fast-LIO2 (MID360) under ROS2, enabling real-time high-precision mapping and localization for a competitive robotic sentry platform
- Implemented point cloud distortion correction and IMU preintegration to mitigate motion distortion and accumulated drift during aggressive maneuvers
- Designed yaw-angle correction using a PYR filter and global relocalization via Transform point cloud registration to facilitate recovery from localization failures
- Deployed A* global path planning with APF-based dynamic obstacle avoidance; built a Gazebo simulation environment for Sim2Real algorithm transfer

SWJTU-Agricultural Inspection Robot - Core R&D of Multi-Sensor Fusion Algorithms

2025-08 ~ 2025-09

- Designed a multi-sensor fusion localization system on ROS1 integrating C16 mechanical LiDAR, IMU, and RTK-GPS, achieving stable 5+ hour operation in large-scale agricultural environments
- Compensated for RTK signal degradation using a vehicle kinematic model with smooth transition between RTK-fixed and dead-reckoning modes (<30 s switching latency)
- Developed a precision visual alignment system for autonomous charging-station docking, achieving end-effector positioning error < 3 cm

INTERNSHIP EXPERIENCE

Momenta (First listed Physical AI company on HKEX) - Camera Service R&D Intern

2026-01 ~ 2026-04

- Developed camera drivers and image processing pipelines (cylindrical projection, inpainting, AVM surround-view stitching) for multiple production vehicle models, ensuring efficient data input to the perception system
- Developed system-level C++ software on Linux/QNX platforms; independently completed the first-vehicle bring-up and bench debugging for the Robocar project on NVIDIA Thor-U
- Contributed to Camera Agent development: workflow-SKILL-ization — integrated with the Feishu developer platform to enable BOT-triggered automated SKILL execution for routine debugging and development;

CERTIFICATE & HONOR

- National Level 2nd Prize, The 3V3 Competition for the RoboMaster 2025 University League Championship 2025.05
- National Level 3rd Prize, The Sentinel Robot team for the RoboMaster 2025 Super Competition 2025.07
- National Level 3rd Prize, The Aerial Robot team for the RoboMaster 2025 Super Competition 2025.07
- Provincial Level 1st Prize, Personal honor of the RoboMaster 2025 University League Championship 2025.04
- Provincial Level 2nd Prize, Personal honor of the RoboMaster 2025 Super (Regional) Competition 2025.06
- Provincial Level 3rd Prize, Personal honor of the RoboMaster 2025 Super (National) Competition 2025.08

HIGHLIGHTS

- Strong hands-on engineering experience with proven ability to deploy algorithms on real-world platforms
- Proficient in C++/Python and Linux system development; skilled in ROS/ROS2 frameworks and hardware-software synchronization
- Adept at synthesizing methodologies and documenting best practices; self-motivated with strong passion for research and engineering
- Experienced in leveraging cutting-edge AI tools (Claude, Codex)